

DYNAMICAL SYSTEMS

MIDTERM PREPARATION

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EXAM PRACTICE QUESTIONS (CH 1–6.4)

Instructions. Read each question carefully to make sure you’ve answered it fully. You must show your work and explain your reasoning to receive full credit. No calculators or devices of any kind are allowed.

- 1) Consider the one-dimensional flow defined by

$$u' = r + u - u^3.$$

- (a) Sketch all the qualitatively different vector fields that occur as the parameter r is varied. Find the value(s) of r at which bifurcations occur, and classify them as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.
- (b) Sketch a bifurcation diagram of fixed points u^* as a function of the parameter r . Be sure to indicate stability on your diagram.

Solution.

□

- 2) Consider the one-dimensional flow defined by

$$u' = r(u - 1) - u^2(u - 1).$$

- (a) Sketch all the qualitatively different vector fields that occur as the parameter r is varied. Find the value(s) of r at which bifurcations occur, and classify them as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.
- (b) Sketch a bifurcation diagram of fixed points u^* as a function of the parameter r . Be sure to indicate stability on your diagram.

Solution.

□

- 3) The following equation describe motion on a line.

$$x' = bx(b - x^2).$$

- (a) Sketch all the qualitatively different vector fields that occur as b is varied. Find any values of b at which bifurcations occur, and classify these as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.
- (b) Sketch the bifurcation diagram of fixed points x^* vs. b , labeling the stability of each fixed point.
- (c) Describe the basin of attraction(s) for the stable fixed point(s) that exist when $b = 1$.

Solution.

□

- 4) The following equation describe motion on a line.

$$\frac{dx}{dt} = (r - 1 + x)(e^x - r).$$

- (a) Sketch all the qualitatively different vector fields that occur as b is varied. Find any values of b at which bifurcations occur, and classify these as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.
- (b) Sketch the bifurcation diagram of fixed points x^* vs. b , labeling the stability of each fixed point.
- (c) Describe the basin of attraction(s) for the stable fixed point(s) that exist when $b = 1$.

Solution.

□

- 5) Consider the system $\dot{x} = ax - x^2 + x^3$.
- (a) Sketch all the qualitatively different vector fields that occur as a is varied. Find the values of a at which bifurcations occur, and classify these as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.
 - (b) Sketch the bifurcation diagram of fixed points x^* vs. a .
 - (c) Can hysteresis occur in this system? Explain why or why not.

Solution. □

- 6) Consider the system $\dot{x} = x - a(x^2 + x)$.
- (a) Sketch all the qualitatively different vector fields that occur as a is varied. Find the value(s) of a at which bifurcations occur, and classify them as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.
 - (b) Sketch the bifurcation diagram of fixed points x^* vs. a .

Solution. □

- 7) For the following system of differential equations,

$$\mathbf{x}' = \begin{bmatrix} 2 & \alpha^2 \\ -1 & -4 \end{bmatrix} \mathbf{x}$$

where $\alpha > 0$ is an unspecified constant,

- (a) determine the eigenvalues of the coefficient matrix in terms of α ,
- (b) find the critical values of α at which the qualitative nature of the phase portrait experiences a dramatic change, and
- (c) sketch the phase portrait for the different choices of α : one just below the each critical value and one just above each critical value.

Solution. □

- 8) Consider the system 2D linear system $\dot{x} = Ax$, where $A = \begin{bmatrix} a & -1 \\ -1 & b \end{bmatrix}$ for parameters $a > 0$ and $b > 0$.
- (a) Sketch a phase portrait and classify the fixed point at the origin for all possible qualitatively distinct cases as a function of parameters. Be sure to include eigenvectors associated with stable/unstable directions in your sketch as appropriate.
 - (b) Summarize these results by indicating the regions that you've identified in the first quadrant of the (a, b) -plane.

Solution. □

- 9) Consider the following planar dynamical system:

$$\begin{aligned} \dot{x} &= y - y^3 \\ \dot{y} &= -x - \frac{1}{2}y^2 \end{aligned}$$

- (a) Find all fixed points and classify them using eigenvalues of the linearized system. For fixed points with real eigenvalues, find the eigenvectors as well.
- (b) Sketch the nullclines and a plausible phase portrait for this system.

Solution. □

- 10) Consider the following planar dynamical system:

$$\begin{aligned} \dot{x} &= x - xy \\ \dot{y} &= -y + xy - y^2 \end{aligned}$$

- (a) Find all fixed points and classify them using eigenvalues of the linearized system. For fixed points with real eigenvalues, find the eigenvectors as well.
- (b) Sketch the nullclines and a plausible phase portrait for this system.

Solution. □

- 11) Consider the system

$$\dot{x} = x^2 - y, \quad \dot{y} = x - y.$$

- (a) Find and classify all fixed points of this system, including eigenvalues and eigenvectors.
 (b) Sketch the nullclines and a plausible phase portrait for this system.

Solution.

□

- 12) Consider the system

$$\dot{x} = x - x^3, \quad \dot{y} = -y.$$

- (a) Find and classify all fixed points of this system, including eigenvalues and eigenvectors.
 (b) Sketch the nullclines and a plausible phase portrait for this system.

Solution.

□

- 13) An ODE system models the evolution of social distancing norms during a pandemic. The model assumes individuals reduce social contacts when infection rates are high and increase social contacts when infection rates are low. The model has two variables: the level of norm adherence in the community (x) and the proportion of infected persons in the community (y). The non-dimensionalized equations are shown below:

$$x' = ay + b - dx$$

$$y' = r(1 - x)y(1 - y) - y$$

where $x \in (0, \infty)$ and $y \in [0, 1]$. Assume $a \geq 1$, $b > 0$, $d > 0$ and $r > 0$.

- (a) Show the model has a unique disease-free equilibrium (DFE) $(x^*, 0)$ with $x^* > 0$.
 (b) Show there is always at least one endemic equilibrium (EE) $(x^*, y^*) \in \mathbb{R}^2$ with $x^*, y^* > 0$

Hint: solve for the non-zero equilibrium in x . This should yield a quadratic equation of the form $\alpha x^2 + \beta x + \gamma = 0$. Analyze the coefficients then state conditions for positive roots.

- (c) Suppose $b < d$. Show that the DFE is locally stable if $r < r_c$. What is r_c ?

Solution.

□

- 14) Consider the 2nd-order ODE that describes the angle ϕ of a bead on a rotating hoop:

$$mr\ddot{\phi} + b\dot{\phi} + mg \sin \phi - mr\omega^2 \sin \phi \cos \phi = 0.$$

Here, m, b, g, r are positive parameters while $\omega \geq 0$.

- (a) By nondimensionalizing, show that the ODE can be written in the form

$$\varepsilon \phi'' + \phi' + \sin \phi - \gamma \sin \phi \cos \phi = 0.$$

for some dimensionless parameters ε and γ . Express ε and γ in terms of the parameters of the original system.

- (b) Under what assumptions about the original parameters can the system be well-approximated by the following 1D flow?

$$\phi' = -\sin \phi + \gamma \sin \phi \cos \phi$$

- (c) Assuming that $\gamma = 0$, find the fixed points of the above 1D flow and sketch a phase portrait.
 (d) Argue that a pitchfork bifurcation occurs at $\gamma = 1$. You may use graphical methods or any other valid approach to support your argument.

Solution.

□

- 15) (25 points) Consider the 1D flow defined by

$$u' = r(u - 1) - u^2(u - 1).$$

- (a) Sketch all the qualitatively different vector fields that occur as the parameter r is varied.
 (b) Find the value(s) of r at which bifurcations occur, and classify them as saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork.
 (c) Sketch a bifurcation diagram of fixed points u^* as a function of the parameter r . Be sure to indicate stability on your diagram.

Solution.

□

16) (25 points) Consider the system

$$\begin{aligned}\dot{x} &= -y \\ \dot{y} &= 2x(1 - x^2)\end{aligned}$$

- (a) Find all fixed points of the system.
- (b) Classify each of the fixed points you found in part (a), using eigenvalues of the linearized system. For fixed points with real eigenvalues, find the eigenvectors as well.
- (c) Show that quantity $E = ax^2 + by^2 + x^4$ is conserved along trajectories for a suitable choice of a and b .
- (d) Argue for the existence of periodic orbits in this model. You may use any valid method to support this claim.
- (e) Sketch the nullclines and a plausible phase portrait for this system.

Solution.

□

17) (25 points) Consider the system 2D linear system $\dot{x} = Ax$, where $A = \begin{bmatrix} a & -1 \\ -1 & b \end{bmatrix}$ for parameters $a > 0$ and $b > 0$.

- (a) Sketch a phase portrait and classify the fixed point at the origin for all possible qualitatively distinct cases as a function of parameters.
Note: Qualitatively accurate phase portraits are sufficient here, it is *not* necessary to compute eigenvectors of A .
- (b) Summarize these results by indicating the regions that you've identified in the first quadrant of the (a, b) -plane.

Solution.

□

18) (25 points) Consider the 2nd-order ODE that describes the angle ϕ of a bead on a rotating hoop:

$$mr\ddot{\phi} + b\dot{\phi} + mg \sin \phi - mr\omega^2 \sin \phi \cos \phi = 0.$$

Here, m, b, g, r are positive parameters while $\omega \geq 0$.

- (a) By nondimensionalizing, show that the ODE can be written in the form

$$\varepsilon\phi'' + \phi' + \sin \phi - \gamma \sin \phi \cos \phi = 0.$$

for some dimensionless parameters ε and γ . Express ε and γ in terms of the parameters of the original system.

- (b) Under what assumptions about the original parameters can the system be well-approximated by the following 1D flow?

$$\phi' = -\sin \phi + \gamma \sin \phi \cos \phi$$

- (c) Assuming that $\gamma = 0$, find the fixed points of the above 1D flow and sketch a phase portrait.
- (d) Argue that a pitchfork bifurcation occurs at $\gamma = 1$. You may use graphical methods or any other valid approach to support your argument.

Solution.

□